

# Anika Bhutani

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## EDUCATION

### University of California: Berkeley

Berkeley, CA

*Bachelor of Science - Electrical Engineering & Computer Sciences* | GPA: 3.9/4.0

May 2029

- **Courses:** Structure and Interpretation of Computer Programming, Data Structures & Algorithms (Java), Principles and Techniques of Data Science, Great Ideas of Computer Architecture, Discrete Mathematics and Probability Theory, Linear Algebra and Differential Equations
- **Organizations:** Business & Software at Berkeley; The Daily Californian: Data & Projects; Asha at Berkeley; Berkeley SkyDeck

## EXPERIENCE

### Outsampler | Python, PyTorch, Google Cloud

Remote

*AI/ML Engineer Intern*

Jan 2026 - Present

- Fine-tuned code LLM on BIRD-SQL via GRPO reinforcement learning with LoRA; improved SQL execution accuracy from 47% to 81%
- Implemented custom reward functions for format compliance and execution correctness using a SQLite-based evaluation pipeline
- Building LLM-powered pipeline that fetches financial news/SEC filings and summarizes and scores them by severity for stock watchlist

### Luel AI | React, Next.js, pdf-lib

Berkeley, CA

*Software Engineer (Contract)*

Feb 2026 - May 2026

- Built end-to-end PDF e-signature flow handling upload, real-time preview, signature embedding, and download with client-side validation
- Evaluated perceptual hashing vs CNN-based frame dimensionality reduction for duplicate submission detection in video/audio training data
- Leveraged Claude and Cursor throughout to accelerate iteration and critically evaluated AI-generated code against production requirements

### PharmaACE | Python, Scikit-Learn, Pandas

Remote

*Data Science Intern*

Jun 2025 - Aug 2025

- Developed cancer risk prediction model using Random Forests, achieving 95% accuracy through feature selection and cross-validation
- Engineered physician referral network from patient-level data using graph-based modeling to analyze referral connectivity patterns

### Stanford University | Python, R, SQL, Java

Palo Alto, CA

*Data Science Research Intern*

May 2024 - Oct 2024

- Contributed to spatial transcriptomics research; built analysis pipelines in Python and R to process high-dimensional gene expression data
- Reduced per-image microscopy processing time by ~83% by developing automated Java macros to replace fully manual image analysis

## PROJECTS

### Smart Clinical | Python, JavaScript, HTML, Tableau

Nov 2025

- Designed hospital resource allocation app integrating a Random Forest risk prediction model to optimize care prioritization per patient
- Built full-stack interface with Streamlit and JavaScript, integrating external APIs; placed 2nd at UC Berkeley's Datathon for Social Good

### Kinkade Theater | SciPY, Python

Apr 2025

- Built a 3D FDTD acoustic simulation in Python modeling wave propagation and boundary reflections in a physics-based discretized theater
- Used SciPy differential evolution to optimize placement of diffusers in Kinkade Theater geometry, benchmarked against acoustics literature

### Energy Equity Model | Python, Pandas

Feb 2025

- Scraped demographic, economic, and irradiance datasets and developed a regression-based machine learning model in Python to predict regional solar energy potential in weak regions; performed feature engineering and data preprocessing in Pandas, achieving an  $R^2$  of 0.92

### Justice Junction | Java, CSS

Apr 2024

- Developed interactive educational game simulating social justice scenarios with real outcome data; 2nd/600 at Western Digital Hackathon

### Pneumonia Detection | Python, Scikit-Learn

Jul 2023

- Implemented KNN and CNN architectures for pneumonia classification on X-ray images, did image preprocessing and train/test evaluation
- Applied data augmentation and tuned hyperparameters to improve model generalization, achieving 86% test accuracy with the CNN model

## LEADERSHIP

### Business and Software at Berkeley

Berkeley, CA

*Project Manager*

May 2026 - Present

- Sourced consulting projects from FAANG and F500 companies through direct outreach and client qualification for a consulting org
- Led team of developers through project lifecycle — scoping deliverables, delegating tasks, running client dev calls, and tracking milestones

### Snowflake Learning

Santa Clara, CA

*Computer Science and Physics Lead*

Feb 2024 - May 2025

- Created AP Computer Science and AP Physics C: Mechanics curricula for students with learning disabilities; increased user base by ~16%

## SKILLS

**Languages, Frameworks, Libraries:** Python, Java, SQL, HTML, Scheme, JavaScript, React, NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn

**Tools & Concepts:** Git, GRPO/RL fine-tuning, LLMs, Cursor, Claude, Jupyter, Tableau, Figma, VS Code, Microsoft Office, API integration